*Teaching Notes 1A**
Present Value

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*Many thanks to Prof. Amit Seru who made his material available to me. These teaching notes closely follow his.

Overview

- How Do We Calculate Value?
  - The NPV Rule.
- Present Value Concepts:
  - Time Value of Money
  - Future Value, Present Value.
  - Value Additivity
  - Annuity Value, Perpetuity Value.
  - Growing Annuity Value, Growing Perpetuity Value
- Examples
- What do we use NPV for?
- Appendix: Methods of Calculation Using Excel

Objectives

- Understand present and future value concepts and their application to a variety of simple valuation problems.

- Understand two key concepts in finance.
  - Time
  - Uncertainty

- Explore the concept of Value Additivity.

- Present Values are Market Prices.

How Do We Calculate Value?

By value we mean the **market value**

(or alternatively, the price)

Crucial Concepts In Finance

- **Time Value of Money:** Cash flows that occur later are worth **less** today.
- **Uncertainty:** Risky cash flows are worth **less** than safe cash flows.
- **Law of One Price:** Two projects that produce "identical" cash flows should have the **same** price or value.
- **Replication of Cash Flows:**
  - If stream of cash flows from an asset over time is the same as cash flows from a project
    - Value of the asset = Value of the project

The NPV Rule

- How do we go about answering the fundamental question, "What projects should the firm take on?"

The firm should invest in a project if taking the project returns a higher value than the costs of undertaking the project.

- That is, a manager should choose projects with a positive net present value:

Net Present Value (NPV) = Project Value - Project Cost

The NPV Rule: Benefits And Estimation

- **Benefits:**
  - Projects are accepted if they increase shareholder value.
  - Allows separation of ownership and control.
  - Simplicity.
- **Estimation:**
  - Net present value requires valuing cash flows:
    - Arriving at different points in the future.
    - With different degrees of uncertainty or risk.

- **Bottom Line:** Accounting for these two problems provides the framework for determining value.
  - Finance is the study of the effect of **time** and **uncertainty** on value.

Present Value Concepts

A dollar today is worth **more** than a dollar next year because it can earn interest (if positive)

Time Value Of Money

- We can move cash flows through time by using market interest rates.
  - \$1 today is worth **\$1.05** in one year if we can lend (invest) at a one-year interest rate of 5% (Future Value).
  - \$1 in a year is worth about **\$0.95** today if we can borrow at a one-year interest rate of 5% (Present Value).

- **Example:** Importance of Time Value of Money – Flu Vaccine
  - **Strategy A:** Bring to market in 1 year. Invest \$1 billion (B) now. Returns \$500 million (M), \$400M and \$300M in years 1, 2 and 3 respectively.
  - **Strategy B:** Bring to market in 2 years. Invest \$200M in years 0 and 1. Returns \$300M in years 2 and 3.

- **Question:** Which strategy creates the most value?

Time Value Of Money

- Our **key** tool in the first part of the course will be the **interest rate**, or equivalently, the **discount rate**; or **cost of capital**; or **opportunity cost of capital**; or

$$\boxed{r}$$

- The interest rate is affected by **time** and **risk**.

- An interest rate expresses the cost of borrowing or the profit from lending on a per-dollar, per-unit of time basis.
  - Unless, otherwise stated, we will assume equal borrowing and lending rates.
- **Example:** In the credit market when we lend $1 for one year, we can get a repayment of $1.05 we say the interest rate is 5%.

Time Value Of Money

- If the loan market is *liquid* and *frictions* are small, then our lending and borrowing rates should be the same whether we borrow $1,000 or $1 million (assuming equal repayment rates!)

- Interest rates could differ depending on the amount **borrowed/lent**:
  - Very relevant for the Federal Government: large deficits financed with borrowing could drive up interest rates
  - And Corporations: Will spend a lot of time thinking about this in the later part of the course

Future Value

- **Definition:** The **Future Value** (**FV**) is what a certain dollar amount today, if invested, will be worth to you at some time in the future.
  - The future value of the cash flow depends both on the amount and on how far into the future you're looking.
- **Example:**
  - Assume you have \$100 and can invest at 8% per year.
  - Value in 1 year is: $\$100 \times (1 + 0.08) = \$108$
  - After 2 years: $\$100 \times 1.08 \times 1.08 = 100 \times 1.08^2 = \$116.64$
  - After n years: $\$100 \times 1.08^n$
- In general the **future value** in n years of *P* invested today is:

$$\mathbf{FV} = P \times (1 + r) \times \ldots \times (1 + r) = P \times (1 + r)^n$$

Future Value

- **Example:**
  - Invest \$3.00 for one period at 10% interest rate.
  - You earn \$0.30 in interest, producing \$3.30 total.
  - $(1 + 0.10) \times \$3.00 = \$3.30$
- **Example:**
  - Invest \$3.00 for two periods at 10% per period. At the end of the first period, you have \$3.30, which then earns 10% interest.
  - $(1 + 0.10) \times \$3.30 = (1 + 0.10)^2 \times \$3.00 = \$3.63$
- **Example:**
  - Invest \$3.00 for three periods at 10% interest per period.
  - $(1.10)^3 \times \$3.00 = \$3.99$

Future Value Factor $\rightarrow (1+r)^n = FV/P$

Present Value

- **Definition:** The **Present Value** (**PV**) of a future cash flow is the amount that you would need to invest today to build up to that future cash flow.
- **Example:**
  - With an interest rate of 6%, what is the PV of \$100 received one year from now?

$$PV \times (1 + 0.06) = \$100 \rightarrow \mathbf{PV} = \$100 / 1.06 = \$94.34$$

- **Example:**
  - With an interest rate of 10%, what is the PV of \$100 received one year from now?

$$PV = \$100/1.10 = \$90.91$$

- **Note:** The **higher** the interest rate, the **lower** the present value.

Present Value

- **Example:**
  - With an interest rate of 10%, what is the Present Value of \$100 received **two years** from now?

$$PV \times (1.10)^2 = \$100 \rightarrow \textbf{PV} = \$100/1.10^2 = \$82.64$$

- **Note:** The **longer** the time until the cash flow, the **lower** the present value.

- **Bottom Line:** In general, with an interest rate of r, and a future cash flow F payable in n years, the **present value** is:

$$PV = \frac{\text{Future Cash } Flow}{(1+r)^n} = F \times \frac{1}{(1+r)^n}$$

Present Value

- **Example:**
  - How much do we need to invest now at 10% to have \$3.30 at the end of one period?
  - We already know the answer is \$3.00; but if we didn't, we'd solve for it this way:

    $$(1.10) \times \boldsymbol{x} = \$3.30 \rightarrow \boldsymbol{x} = \$3.30/(1.10) = \$3.00.$$

    - \$3.00 is the present value of \$3.30 at "time 1" when the interest rate is 10%.
  - \$3.00 is the present value of \$3.99 at "time 3" when the interest rate is 10%, which we could compute as:

    $$PV = \$3.99/(1.10)^3 = \$3.00$$

Present Value Factor → $1/(1+r)^n = PV/F$

Value Additivity

- The present value of a stream of future cash flows is simply the sum of the present values of the individual cash flows.

- Consider the stream of future cash flows:
  - \$3.30 at $t = 1$, \$3.63 at $t = 2$, \$3.99 at $t = 3$.
- The present value of this stream (at 10%) is the amount you need to invest today to produce the stream.
  - We already know that an investment of \$3.00 produces each of the components of the stream.
  - So to produce the entire stream would require an investment of \$9.00, which is the present value of the stream of cash-flows.

Present Value $\rightarrow$ $PV = C_1/(1+r) + C_2/(1+r)^2 + \ldots + C_T/(1+r)^T$

Value Additivity

- **Example:** Retirement Account
  - Starting 1 year from today, deposit $2000/year in a retirement account. If *r* = 6%, how much will you have in 30 years?
  - Key to solve this problem is to move each cash flow to the same point in time and only then add.

Year	Deposit	Value in 30 years
1	$2,000	$\$2{,}000 \times (1.06)^{29} = \$10{,}836.78$
2	$2,000	$\$2{,}000 \times (1.06)^{28} = \$10{,}223.37$
[…]	[…]	[…]
30	$2,000	$2,000
	Total	**$158,116.37**

- However, in a minute we'll see that there is a formula for simple **annuities** like this one.

Value Additivity

- **Example:** Flu Vaccine (Continue From Earlier Slide)
  - Assume that the interest rate is 5%.

Strategy A				
Time	**0**	**1**	**2**	**3**
Cash Flows	-1000	500	400	300
PV	-1000	476.19	362.81	259.15
			NPV	**98.15**

Strategy B				
Time	**0**	**1**	**2**	**3**
Cash Flows	-200	-200	300	300
PV	-200	-190.48	272.11	259.15
			NPV	**140.78**

- The firm should choose strategy B, since NPV(B) > NPV(A).

Annuity Value and Perpetuity Value

- There are a number of series such as annuities and perpetuities that are of practical importance
  - Will be using them heavily in our calculations of firm value
- However, there are only three formulas that you need to remember. The rest are simple extensions.
  1. Present and Future Value of a single cash ($1) cash flow:

     $PV = 1/(1 + r)^n$ and **$FV = (1 + r)^n$**

  2. Present Value of a Unit Perpetuity: **$PV = 1/r$**
  3. Present Value of a Unit *Growing* Perpetuity: **$PV = 1/(r - g)$**
- Let's go over the perpetuity and growing perpetuity formulas, and show how to get all of the other formulas you need from these.

Annuity Value and Perpetuity Value (cntd)

- Useful short-cut formula to use when cash-flows are all the same or grow at a constant rate, indefinitely.
  - Even though the sum of the payments of a perpetuity is infinite, the value is not.
  - As long as the interest rate is positive, the discounted value of payments will be finite.
- A **perpetuity** is an infinite stream of identical cash flows, C.
  - Using the formula for the sum of a geometric series, it turns out that:

PV(Perpetuity) = C/r

  - It is important to note that this formula is correct only if the first payment arrives at $t = 1$.
  - **Question:** What is the formula if it arrives at $t = 0$?

Annuity Value and Perpetuity Value (cntd)

- A perpetuity's value is the sum of the present value of the payments from holding the perpetuity forever:

$$P = C/(1 + r) + C/(1 + r)^2 + C/(1 + r)^3 + \ldots$$

- Alternately, suppose you bought the perpetuity today and sold it for P_1 in a year. Then the value of the perpetuity is the present value of the cash flow $C + P_1$ you will receive next year:

$$P = 1/(1 + r) \times [C + P_1]$$

- *But the present value of the perpetuity must be the same at each point in time.* Why?
  - Because at each point in time the future cash flows are identical. Therefore $P = P_1$

$$P = 1/(1 + r) \times [C + P] \rightarrow \quad P = C/r$$

Annuity Value and Perpetuity Value (cntd)

- A **growing perpetuity** is an infinite stream of cash flows that grow at a constant rate (g).
  - Cash Flow Stream is:

    $$C, \quad (1 + g)\times C, \quad (1 + g)^2\times C, \quad (1 + g)^3\times C, \ldots$$

  - Using the formula for the sum of a geometric series, it turns out that:

    $$\textbf{PV(Growing Perpetuity)} = \mathbf{C/(r - g)}$$

  - It is important to note that this formula is correct only if the first payment arrives at $t = 1$.
  - **Question:** What is the formula if it arrives at $t = 0$?

Annuity Value and Perpetuity Value (cntd)

- The **growing perpetuity** value is the sum of the present value of the payments from holding the perpetuity forever:

$$P = C/(1 + r) + C(1 + g)/(1 + r)^2 + C(1 + g)^2/(1 + r)^3 + \ldots$$

- Alternately, suppose you bought the perpetuity today and sold it for P_1 in a year. Then the value of the perpetuity is the present value of the cash flow $C + P_1$ you will receive next year:

$$P = 1/(1 + r) \times [C + P_1]$$

- *The price in 1 year must be (1 + g) times its price now.* Why? Because it's future cash flows are all g% larger.
  - Therefore $P \times (1 + g) = P_1$

$$P = 1/(1 + r) \times [C + P \times (1 + g)] \rightarrow P = C/(r - g)$$

Annuity Value and Perpetuity Value (cntd)

- An **annuity** is a *finite* stream of constant payments.
  - It is a fixed payment starting in 1 year and lasting for n years.
  - If the first cash flow, C, arrives at t = 1 and the last arrives at time n, the present value of an annuity is:

$$PV(Annuity) = \frac{C}{r} \times \left(1 - \frac{1}{(1+r)^n}\right)$$

  - Note that the cash flow stream to an annuity can be constructed by taking the stream to a perpetuity that starts at time 1, and subtracting off the stream to a perpetuity that starts at time n + 1.

$$PV(Annuity) = \frac{C}{r} - \frac{C/r}{(1+r)^n}$$

Annuity Present Value Factor → $\frac{1}{r} \times \left(1 - \frac{1}{(1+r)^n}\right) = P/C(r,n)$

Annuity Value and Perpetuity Value (cntd)

- A **growing annuity** is a finite stream of cash flows whose payments grow at a constant rate (g).
  - Cash Flow Stream is: C, (1+g)C, $(1+g)^2$C, $(1+g)^3$C, ... ,$(1+g)^{n-1}$C

- Compute the growing perpetuity as the difference between the present values of two growing perpetuities with growth rate g.

$$PV(Growing\ Annuity) = \frac{C}{r-g} - \frac{\frac{C(1+g)^n}{r-g}}{(1+r)^n}$$

$$= \frac{C}{r-g} \times \left(1 - \frac{(1+g)^n}{(1+r)^n}\right)$$

- Annuities show up all the time in real life.
  - Every type of loan has an annuity component.

Examples

Example I: Annuity and Future Value

- What is the present value of \$5,000 per year, starting in one year, for 7 years? Assume r = 11.75%.

$$\$5{,}000 \times P/C(11.75\%, 7) = \frac{\$5{,}000}{0.1175} \times \left(1 - \frac{1}{(1+0.1175)^7}\right) = \$23{,}000.82$$

- What is the future value in year 7?

$$\$5{,}000 \times P/C(11.75, 7) \times (1+0.1175)^7 = \frac{\$5{,}000}{0.1175} \times \left(1 - \frac{1}{(1+0.1175)^7}\right) \times (1+0.1175)^7$$
$$= \$50{,}058.30$$

- **Note:** See **Appendix** for Methods of Calculation using Excel Spreadsheets and Financial Calculators.

Example II: Annuity Contract and Car Loan

- **Annuity Contract:** An insurance company is selling an annuity of \$10,000 per year for 20 years. What is the fair market price or present value if r = 5%?

$$PV = \frac{C}{r} \times \left(1 - \frac{1}{(1+r)^n}\right) = \frac{\$10{,}000}{0.05} \times \left(1 - \frac{1}{(1.05)^{20}}\right) = \$124{,}622.1$$

  - [Excel: PV(r, NPER, PMT, FV, Type) = PV(0.05, 20, 10000, 0, 0)]

- **Car Loan:** A car loan of \$20,000 is financed at a 12% annual interest rate over 5 years. What are the annual payments?

$$PV = \frac{C}{r} \times \left(1 - \frac{1}{(1+r)^n}\right) \to \$20{,}000 = \frac{C}{0.12} \times \left(1 - \frac{1}{(1.12)^5}\right) \to C = \$5{,}548.19$$

  - [Excel: PMT(r, NPER, PV, FV, Type) = PMT(0.12, 5, 20000, 0, 0)]

Example III: Bank Deposit

- How much will you have in the bank in 5 years if, starting in a year, you deposit $1,000 each year and you earn 5% per year?

$$PV = \frac{C}{r} \times \left(1 - \frac{1}{(1+r)^n}\right)$$

$$FV = PV \times (1+r)^n = \frac{1{,}000}{.05} \times \left(1 - \frac{1}{(1.05)^5}\right) \times (1.05)^5 = \$5{,}525.63$$

- [Excel: FV(r, NPER, PMT, PV, Type) = FV(0.05, 5, -1000, 0, 0)]

Example IV: Pension Planning

- A pension fund manager must fund a $10M obligation due in 10 years. What annual contributions are needed if r = 5%?

$$PV = \frac{C}{r} \times \left(1 - \frac{1}{(1+r)^n}\right)$$

$$FV = \$10M = PV \times (1+r)^n = \frac{C}{.05} \times \left(1 - \frac{1}{(1.05)^{10}}\right) \times (1.05)^{10}$$

$$\rightarrow C = \$0.795M$$

- The pension fund manager must invest $795,000 per year.
- [Excel: PMT(r, NPER, PV, FV, Type) = PMT (0.05, 10, 0, 10000000, 0)]

Example V: Paying Off A Loan

- A entrepreneur borrows \$300,000 today. The interest rate is 8%. If the entrepreneur makes annual payments of \$45,000 per year how many years will it take to repay the loan?

$$PV = \frac{C}{r} \times \left(1 - \frac{1}{(1+r)^n}\right) \rightarrow \frac{1}{(1+r)^n} = \left(1 - \frac{PV \times r}{C}\right)$$

$$\rightarrow ln\left(\frac{1}{(1+r)^n}\right) = ln\left(1 - \frac{PV \times r}{C}\right)$$

$$\rightarrow ln(1) - n \times ln(1+r) = ln\left(1 - \frac{PV \times r}{C}\right)$$

$$\rightarrow n = -\frac{ln\left(1 - \frac{PV \times r}{C}\right)}{ln(1+r)}$$

$$= -\frac{ln\left(1 - \frac{300{,}000 \times .08}{45{,}000}\right)}{ln(1+.08)} = 9.90\ years$$

- [Excel: NPER(r, PMT, PV, FV, Type) = NPER(0.08, -45000, 300000, 0, 0)]

Example VI: Planning For Retirement

- Suppose it is your 25th birthday. You will be able to save for the next 25 years, until age 50, i.e., payments on birthdays 26 through 50, 25 in all. Between age 50 to 60, your income will just cover your expenses. Finally, you expect to retire at age 60 and live until age 80 (20 years of retirement).
  - Suppose that you want to guarantee yourself a supplemental income of $30,000 per year after retirement (withdrawals on birthdays 61 to 80, 20 in all).
  - Assume that the interest rate is r = 12%.

- **Question:** How much should you put away every year, for the next 25 years, starting at the end of this year?

Example VI: Planning For Retirement (cont.)

- We should divide the problem:
  - **A:** How much will you need at age 60?
  - **B:** How much will you need at age 50?
  - **C:** How much should you put away every year for 25 years?

Example VI: Planning For Retirement (cont.)

A. The answer is the present value of a 20 year, \$30,000 annuity.

$$PV(Annuity)_{60} = \frac{C}{r} \times \left(1 - \frac{1}{(1+r)^n}\right) = \frac{30{,}000}{0.12} \times \left(1 - \frac{1}{(1.12)^{20}}\right) = \$224{,}083.31$$

B. The present value of the sum that is needed at age 60. Thus, we simply take the present value of P_{60} for 10 years.

$$\mathrm{PV}_{50} = \frac{F}{(1+r)^n} = \frac{224{,}083.31}{(1+0.12)^{10}} = \$72{,}148.83$$

C. An annuity that yields a future value of PV_{50} in 25 years.

$$\frac{C}{r} \times \left(1 - \frac{1}{(1+r)^n}\right) \times (1+r)^n = \$72{,}148.83$$

$$\rightarrow C = \frac{\dfrac{\$72{,}148.83 \times 0.12}{(1+0.12)^{25}}}{\left(1 - \dfrac{1}{(1+0.12)^{25}}\right)} = \$541.11$$

Example VII: Stock Price

- Super Growth Incorporated will pay an annual dividend next year of $3. After analyzing the company you expect that after that the dividend will grow at the rate of 5% per year forever. Also for the companies of this risk class, you calculate the opportunity cost of capital is 10%.

  - **Question:** What should Super Growth's price per share be?

- This is an example of a growing perpetuity:

$$PV = \frac{3}{1.10} + \frac{3(1+.05)}{1.10^2} + + \frac{3(1+.05)^2}{1.10^3} + \cdots = \frac{3}{0.10-.05} = \$60$$

Example VIII: Capital Budgeting

- Consider the case of the purchase of a machine that costs today \$5,000, with running costs equal to \$2,000 which generates revenues equal to \$5,000 (forever). If the interest rate is 10%, calculate the NPV of the investment.

Time	**0**	**1**	**2**	**3**	**4**	**...**	**Forever**
Capital Expenditures	-5000						
Running Costs		-2000	-2000	-2000	-2000	...	-2000
Revenues		5000	5000	5000	5000	...	5000
Cash Flows	**-5000**	**3000**	**3000**	**3000**	**3000**	**...**	**3000**
						NPV	**25000**

$$NPV = -5000 + \frac{3000}{0.10} = 25000$$

- **Question:** How would the result change if after t = 1 the costs and revenues grow at an equal rate of 2% forever?

$$NPV = -5000 + \frac{3000}{0.10 - 0.02} = \$32{,}500$$

Example IX: Retirement Savings Contract

- An insurance agent approaches you and would like to sell you the following retirement savings contract:
  - You pay \$5,000 per year for the coming 15 years.
  - In return, you will receive \$7,000 a year for the following 15 years.
  - Assume that interest rates will remain at constant 9%.
  - **Question:** Should you buy this contract?
- The $\text{NPV}_{t=0}$ of this contract is:

$$NPV = -\frac{5{,}000}{1.09} - \frac{5{,}000}{1.09^2} - \cdots - \frac{5{,}000}{1.09^{15}} + \frac{7{,}000}{1.09^{16}} + \frac{7{,}000}{1.09^{17}} + \cdots + \frac{7{,}000}{1.09^{30}} = ?$$

- You can just compute this NPV the "manual" way, or you can just divide the problem. How?

Example IX: Retirement Savings Contract (cont.)

- In 15 years, what will be value of the annuity?

$$PV(Annuity)_{15} = \frac{C}{r} \times \left(1 - \frac{1}{(1+r)^n}\right) = \frac{7{,}000}{0.09} \times \left(1 - \frac{1}{(1.09)^{15}}\right) = \$56{,}424.82$$

- What is its value today?

$$PV(Annuity)_0 = \frac{\$56{,}424.82}{(1+.09)^{15}} = \$15{,}490.76$$

- What is the time 0 present value of the payments?

$$PV(Payments)_0 = \frac{C}{r} \times \left(1 - \frac{1}{(1+r)^n}\right) = \frac{5{,}000}{0.09} \times \left(1 - \frac{1}{(1.09)^{15}}\right) = \$40{,}303.44$$

- The Net Present Value is the difference between the present value of the received annuity and the payments:

$$NPV = \$15{,}490.76 - \$40{,}303.44 = -\$24{,}812.68$$

  - This is the profit to the insurance company measured in today's dollars! (or equivalently our loss).

Example IX: Retirement Savings Contract (cont.)

- Suppose you wanted to bargain to receive a higher annual payment than $7,000.
  - **Question:** At what annual payment would the deal generate zero profits for the insurance company?

$$PV(Payment)_0 = PV(Annuity)_0$$

$$\$40{,}303.44 = \frac{C}{0.09} \times \left(1 - \frac{1}{(1.09)^{15}}\right) \times \frac{1}{(1.09)^{15}}$$

$$\rightarrow C = \$18{,}212.41$$

  - **Note:** You could use the "Solver" function in Excel.

What do we use NPV for?

Why Do We Use The NPV Rule?

- Many reasons for using the NPV rule which will be clearer as the course progresses.

- Suppose a new project has an NPV = $138. The firm's market capitalization is $ 1,000 and the price per share is $1.
  - **Question:** What happens to share price when project is undertaken?

- Share price will rise from the positive NPV addition!
  - Market Capitalization from $1,000 to $1,138
  - Share price from $1 to $1,138/$1,000 = $1.14

- But this does **not** always happen….**Why?**

Acquiring Firms

33,292 views | Jun 16, 2017, 09:52am

Amazon Is Buying Whole Foods For $13.7 Billion

Lauren Debter Forbes Staff

I write about the world's most successful entrepreneurs.

Amazon said on Friday it has agreed to buy Whole Foods in a deal valued at $13.7 billion, as the e-commerce juggernaut makes its boldest push yet into the grocery business.

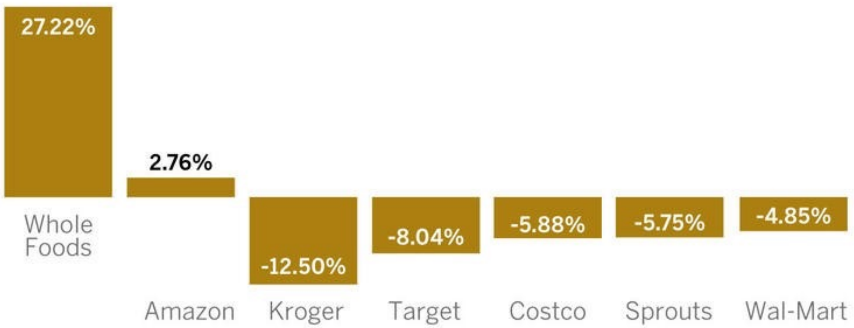
Under the terms of the deal, Amazon would pay $42 per share in cash for the company, which represents a 27% premium to Thursday's closing price.

"Whole Foods Market has been satisfying, delighting and nourishing customers for nearly four decades – they're doing an amazing job and we want that to continue," said Amazon's billionaire founder and CEO Jeff Bezos in a statement.

Shares of Amazon rose 3% on Friday morning, while shares of Whole Foods surged 27%. Grocery companies were reeling on the news that the fierce, low-cost competitor was getting into the space, with stocks like Kroger, Sprouts and Wal-Mart dropping sharply.

Market reaction

Early stock market reaction to Amazon's purchase of Whole Foods Markets Inc.



Source: Google Finance. Data as of 8:52 a.m. PDT @latimesgraphics

Forbes; June 16, 2017

Acquiring Firms

- In this case, the investment project is the acquisition of Whole Foods.
  - Money to undertake a project is raised by issuing debt and/or equity.
  - Amazon issued $16 billion of debt in August 2017 to fund the Whole Foods acquisition.

- The market decides whether this is a positive NPV by looking at future cash flows relative to the money you pay today.

- **Bottom Line:** Shares went up! It meant the market thinks the NPV is **positive**!

New Investments

Nike Shares Close at Another Record High After Controversial Colin Kaepernick Endorsement

Investors have weighed in on Nike's ad campaign that features controversial former NFL quarterback Colin Kaepernick. Despite all of the swooshes scissored away by the company's critics, Nike's stock closed at a record high on Friday.

Nike shares closed up 2 cents to $83.49, the sixth straight day of closing all-time highs. Nike initially stumbled after the public backlash to the endorsement deal with Kaepernick, but its stock has risen 5% since it announced the marketing partnership.

Kaepernick took a knee while the national anthem played during a football game in 2016, protesting the way some police officers treated African-Americans. Donald Trump has kept the controversy at full boil with his tweets on the subject, calling it an insult to the flag.

Fortune; September 14, 2018

In this case, the investment project is the launch of a new advertising campaign.

The cost includes Kaepernick's contract, the creation of new ads, and the airing of these new ads.

The cost has to be compared to the increase in sales revenues to determine whether the project is positive NPV.

Shutting down Projects

Caterpillar's latest restructuring move could cut 880 jobs

Rajesh Kumar Singh 4 MIN READ

CHICAGO (Reuters) - Caterpillar Inc (CAT.N) will close two facilities in Texas and Panama and is also considering shutting its engine manufacturing plant in Illinois as part of a strategy to boost profitability and better handle business cycles, but the move could cut 880 jobs.

The world's largest heavy-duty equipment maker emerged last year from the longest downturn in its history, when sales dropped more than 40 percent between 2012 and 2016.

Chastised by the sales slump, the Deerfield, Illinois-based company has embarked upon a restructuring strategy, looking to squeeze more production out of its existing factories, focusing on lean manufacturing, margin expansion and asset efficiency.

The company's shares were up 1.6 percent at $156.99 on Friday afternoon.

Reuters, March 16, 2018

In this case, Caterpillar announces the closure of two factories and the share price increases.

Question: What does it mean? The NPV of the two factories was **negative**!

Restating Earnings

GE dives after saying it plans to restate 2 years of earnings (GE)

Kimberly Chin
Feb. 26, 2018, 11:30 AM

SHARE

Shares of **General Electric** fell to their lowest level since July 7, 2010 on news that the company will **restate its last two years of earnings** due to the adoption of a new accounting standard, **regulatory filings showed**.

The company said its revised earnings report will likely show earnings per share that are lower than previously stated because the revised accounting standard will now include revenue from long-term contracts. The revised earnings report is estimated to result in a $0.13 drop in its earnings per share for 2016, and a $0.16 cut for 2017, **Reuters reported**.

GE is adopting new accounting standards following a Securities and Exchange Commission investigation into the the company's accounting of long-term service contracts. The move was precipitated by **a recent insurance charge** made by the company, which put regulators on notice.

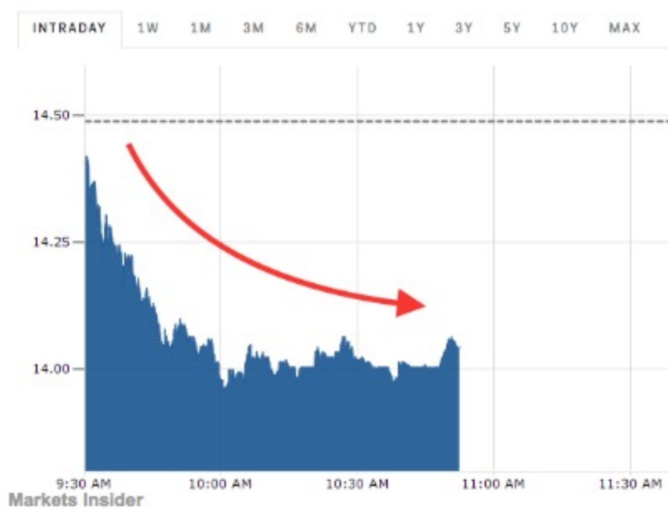
The company **recently took a $6.2 billion hit** on its insurance business after a review of GE Capital's portfolio.

A **Negative** NPV transaction!

GENERAL ELECTRIC (GE)

14.04 USD -0.45 (-3.11%) 10:52:09 AM EST BTT

Prev. Close	14.49	Market Cap (USD)	125.99 B	Day Low 13.96
Open	14.56	Volume (Qty.)	2,027,386	14.06



Business Insider, February 28, 2018

Value of a Manager

Brian Krzanich Out As Intel CEO, Shares Fall 2% Amid Shakeup

Sarah Hansen Contributor

Intel CEO Brian Krzanich has resigned following an investigation into a "past consensual relationship with an Intel employee," the company said in a statement on Thursday. Robert Swan, the company's chief financial officer, will step in as interim CEO.

"An ongoing investigation by internal and external counsel has confirmed a violation of Intel's non-fraternization policy, which applies to all managers," Intel said.

Shares of Intel have fallen roughly 2% since markets closed on Wednesday and were trading at $52.40 at 9:45 on Thursday morning. Shares of the chipmaker have climbed steadily since August 2017, gaining more than 40% after holding steady at around $35 earlier last year.

Krzanich was appointed CEO in May of 2013 after joining Intel in 1982 as an engineer. He previously served as executive vice president and chief operating officer. In recent years, Krzanich has focused on steering Intel towards more "data-centric" businesses, according the company's most recent annual report.

Forbes, June 21, 2018

In this case, the CEO of Intel resigned and Intel's share price immediately declined.

Question: How does the CEO relate to the NPV of future projects?

Toyota loses $1.2bn in value five minutes after Donald Trump's tweet

Shares plummeted after the president-elect vowed to stop the car manufacturer from moving abroad

Rachael Revesz New York | @RachaelRevesz | Thursday 5 January 2017 20:57 | 55 comments

Toyota shares plummeted within minutes of a negative tweet from the president-elect, causing the company to lose around $1.2 billion in value.

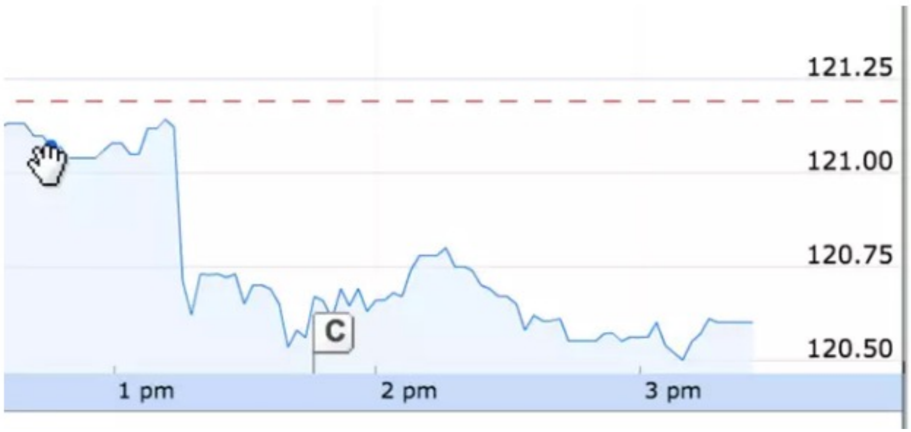
The car company tanked on the stock market within five minutes of Donald Trump vowing to stop the firm from moving to Baja California in Mexico, where it already has a plant.

Mr Trump was confused about the location of the new plant. Toyota had first announced plans for a new plant in Mexico in April 2015 - but in Guanajuato city.

Mr Trump responded there was "no way" the move was going to happen.

"Build plant in US or pay big border tax!" he tweeted.

Mr Trump sent the tweet at 1.14pm ET, and within minutes the share price of the $200 billion market capitalisation company plummeted. Shortly before New York stock markets closed on Thursday, the share price had not recovered and were down from pre-tweet by 0.5 per cent.



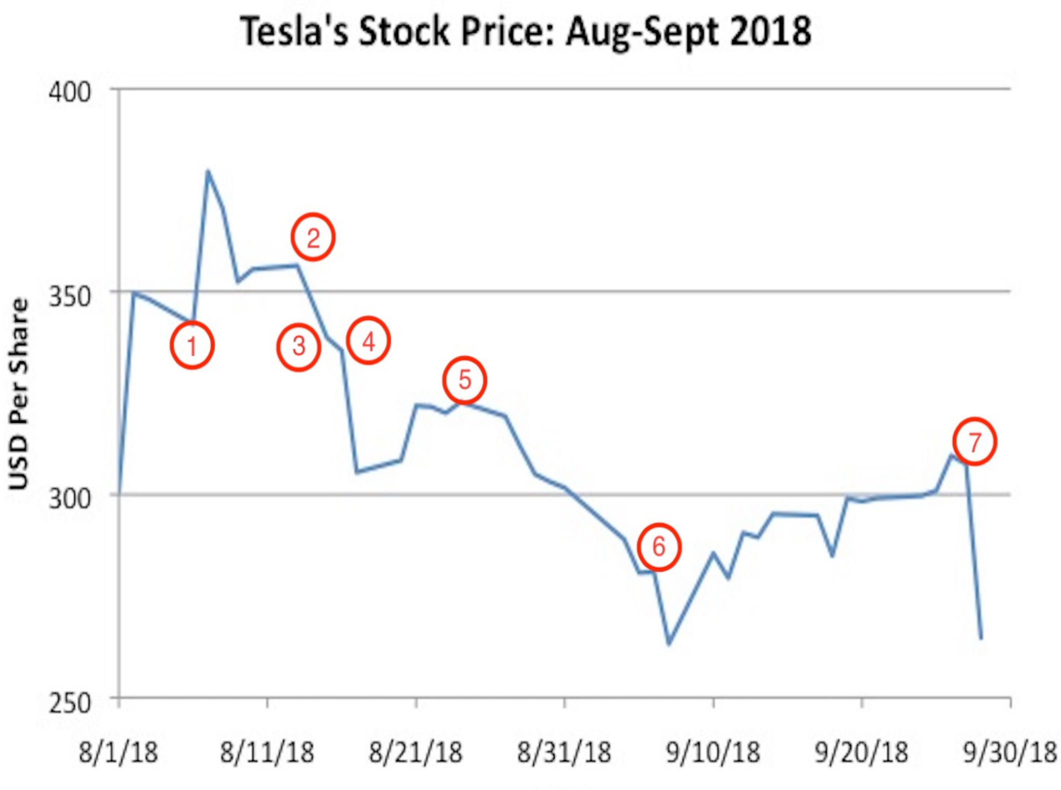
The Independent, January 5, 2017

Politics

In this case, US President-elect Donald Trump announced his intention to block Toyota's new investment project. Toyota's stock price declined immediately.

Question: What does this mean about the NPV of Toyota's proposed project?

Tesla



USA Today, September 28, 2018

1. Aug 7: Elon Musk tweets intent to take Tesla private at $420.
2. Aug 13: Musk tweets he secured funding.
3. Aug 15: Reports of SEC subpoena
4. Aug 16: Musk gives interview about his declining health
5. Aug 24: Musk announces Tesla will remain public
6. Sept 7: Viral video of Musk smoking marijuana
7. Sept 27: SEC accuses Musk of securities fraud

Tesla provides examples of:

(1) acquisition; and

(2) the value of a manager.

Question: How do these events relate to NPV?

Takeaways

- Asset and Market prices are present values.

- **Key Concepts:**
  - **Time Value of Money**: A dollar today is worth **more** than a dollar next year because it can earn interest.
  - **Uncertainty**: Risky cash flows are worth **less** than safe cash flows.
  - **Law of One Price:** Two projects that produce "identical" cash flows should have the **same** price or value.

- **Definitions:**
  - Net Present Value (NPV) = Project Value – Project Cost
  - A **perpetuity** is an infinite stream of identical cash flows.
  - A **growing perpetuity** is an infinite stream of cash flows that grow at a constant rate.
  - An **annuity** is a finite stream of constant payments.

Takeaways (cont.)

- **Definitions:**
  - A **growing annuity** is a finite stream of cash flows whose payments grow at a constant rate.
  - The present value of a stream of future cash flows is simply the sum of the present values of the individual cash flows.
    - This is the concept of **value additivity**.
- NPV used for making corporate investment decisions

Appendix :

Methods of Calculation Using Excel

Remarks About Excel

- **Five** Buttons
  - n = Number of Payments
  - r = Interest Rate Per Payment Period
  - PV = (-) Present Value or Principal
  - PMT = Constant periodic payment amount
  - FV = Future Value or Face Value

- Excel uses the following equation to work with annuities:

$$0 = \text{PV} + \frac{\text{PMT}}{(1+r)} + \frac{\text{PMT}}{(1+r)^2} + \ldots + \frac{\text{PMT}}{(1+r)^{\text{n}}} + \frac{\text{FV}}{(1+r)^{\text{n}}}$$

- **Remark:**
  - Type is the number 0 or 1 and indicates when payments are due.
    - Type = 0 (or omitted): At the end of the period.
    - Type = 1: At the beginning of the period.

Remarks About Excel (cont.)

- Feed in 4 of the inputs and it calculates the other!
  - This is done using the Excel functions:
    - PV: Returns the Present Value for an investment.
    - FV: Returns the Future Value for an investment.
    - PMT: Calculates the payment for an investment.
    - NPER: Returns the number of periods for an investment.
    - RATE: Returns the interest rate per period of an annuity.

- The financial functions in all standard spreadsheets work on the same principle:
  - PV(n, r, PMT, FV) gives the present value
  - FV(n, r, PMT, PV) gives the future value.

- Excel does not have functions for growing annuities, perpetuities, or growing perpetuities. Type the relevant formula!